

Recall-Bounded Durability for Graph-Based Approximate Nearest Neighbor Indexes

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Abstract

Approximate nearest neighbor (ANN) search is evaluated almost entirely on two axes: recall and queries per second. Durability—whether an acknowledged write survives a crash or power loss—is largely absent from the ANN research literature and from the mainstream ANN benchmarks, and is unquantified even in systems that provide it. This paper makes durability a first-class axis for graph ANN. Its foundation is a reusable protocol for measuring durability honestly and a characterization of its cost on one host; its headline is *recall-bounded durability*, which expresses and bounds a crash’s damage as top- k membership at risk and couples that bound to WAL hardening—a new ANN-specific design (the storage mechanisms we apply, group commit and snapshot recovery, are textbook). Using a from-scratch C++20 vector database with a segmented, write-ahead-logged (WAL) HNSW store—whose index reaches 0.999 recall@10 on SIFT1M (and 0.98 on the harder 960-d GIST1M) and sits on the hnsplib recall-QPS Pareto front, confirming we measure on a competitive, not a broken, index—we contribute the following, measured on two hosts (Apple M4 Pro; AWS Linux/NVMe).

(1) A *durability-aware measurement protocol* for ANN indexes: a tmpfs (RAM-backed filesystem) guard, an `fsync`-floor baseline, and mandatory `fsync`-mode labeling, because on some platforms the default `fsync` does not flush the drive cache and so reports optimistic “durable” numbers. (2) An illustration of that protocol catching an `fsync`-mode trap: on macOS/APFS, where plain `fsync` is a page-cache no-op, switching to honest per-write durability (`F_FULLFSYNC`) lowers the device’s durable-sync ceiling by $\sim 155\times$ (49,199 to 317 syncs/s) in a tight-loop microbenchmark—a cautionary example of how mislabeled numbers arise. We confirm on a second host (AWS Linux/NVMe, where plain `fsync` is genuinely durable at 527/s) that this gap is a macOS/APFS artifact while the tax, group commit ($4.9\times$), and recovery ($128\times$) all reproduce—so the findings are not platform-specific. (3) An illustrative durability-tax and group-commit characterization on the SIFT-scale ($d=128$) index: honest per-write durability cuts insert throughput $\sim 4.5\times$ (442 to 98 inserts/s), and a group-commit *batch-size sweep* then recovers the tax along a curve—from $1\times$ at batch 1 to $4.3\times$ at batch 2,000 under `F_FULLFSYNC`, while under plain `fsync` batching barely helps ($\sim 1\times$), showing the speedup is a function of both batch size and `fsync` mode rather than a single constant—with crash-consistency validated both by a fork-and-crash test and by true power-loss injection (Linux `dm-log-writes` replayed to every flush barrier). (4) An illustrative recovery comparison: snapshot-load of a sealed segment is near-constant while WAL replay grows with N (a $5\times$ to $32\times$ gap over $N=1\text{k}–6\text{k}$, widening to $\sim 157\times$ at $N=20\text{k}$); the snapshot-creation cost we measure (tens of ms) is far below the replay it saves, so sealing amortizes; positioned against P-HNSW’s persistent-memory recovery analysis. (5) *Recall-bounded durability*: we express the exposure of an un-durable insert window as top- k membership at risk and bound it (worst case $\min(W, k)/k$; exchangeable expectation W/N), measure on SIFT1M that it is only 0.13% at our batch size ($\sim 770\times$ below the worst case), and prototype a recall-

aware committer whose crash loss we validate against an exact oracle (benign $\Delta\text{recall} \leq \varepsilon$ on SIFT1M); production WAL integration is future work.
 All code, harnesses, and results are open and reproducible; durability numbers are the *median of 7 trials* (3 for the N -sweeps) with the observed min–max range reported alongside.

1 Introduction

Graph-based ANN indexes—Hierarchical Navigable Small World (HNSW) graphs [1] and their descendants—dominate in-memory vector search, and vector databases built on them now back retrieval-augmented generation and semantic search at scale. The community evaluates them on *recall@k* (the fraction of a query’s true k nearest neighbors the index returns) and *throughput* (queries/s, QPS), summarized by the recall-vs-QPS Pareto frontier popularized by ANN-Benchmarks [13] and the Big ANN challenges [14]. What these evaluations almost never measure is *durability*: does an acknowledged write survive a crash?

This is not a niche concern. Vector databases are increasingly operational stores that ingest and update continuously, and an operational store that loses acknowledged writes is not a database. Yet mainstream ANN libraries exclude durable storage entirely (Faiss offers manual serialization, no WAL [10]; hnswlib offers soft-delete tombstones and no crash consistency [2]), and none of the widely used ANN benchmarks (ANN-Benchmarks [13], Big ANN [14], VectorDBBench [15]) include a crash-recovery, **fsync**, or durability case. Production vector databases (Milvus [12], Qdrant, Weaviate) do implement WAL-based persistence, but publish no isolated **fsync**-tax, crash-consistency, or recovery-time analysis. The closest research systems either integrate ANN into an operational DB and *inherit* durability from the host engine—delegating rather than analyzing it [9]—or, like LSM-VEC [8], build an LSM-tree graph index yet report no **fsync** cost, crash consistency, or recovery time.

Our stance is deliberately measurement-oriented: build durability into a *standalone* graph-ANN index using well-known techniques, then measure it honestly and report where the naive measurement misleads. We do not claim the *storage* mechanisms as novel—group commit and snapshot-based recovery are textbook database techniques—and the main contribution is the measurement protocol and the (single-host, median-of-7-trial) characterization it produces. The one genuinely new mechanism is conceptual and ANN-specific: expressing and bounding a crash’s damage as *top- k membership at risk* and coupling it to WAL hardening (§6), which only makes sense because the index is approximate. Two observations motivate the work. First, durable writes are expensive in a way that ANN evaluation has never quantified, and the expense is easy to hide: on macOS (and any stack where **fsync** does not force a drive-cache flush) the “durable” number is a page-cache number. Second, once that cost is paid, the standard techniques apply, but their effect on an ANN index (where each insert also mutates a proximity graph) has not been reported.

Contributions.

- **C1 (§3).** A durability-aware measurement protocol for ANN: tmpfs detection, an **fsync**-floor baseline, and **fsync**-mode labeling, so ANN write/durability numbers are comparable and honest. This is the paper’s primary, reusable contribution.
- **C2 (§4).** An illustrative ANN-isolated durability-tax measurement—throughput and latency percentiles vs. the strength of the durability guarantee—applying the protocol on one host, presented as an example of what it surfaces rather than a general result.

- **C3 (§5).** A group-commit *batch-size sweep* with a fork-and-crash consistency check, showing the recovered throughput is a curve in batch size and depends on `fsync` mode—not a single headline speedup.
- **C4 (§7).** A recovery comparison (snapshot-load vs. WAL-replay) for a standalone graph ANN on conventional block storage (`fsync`/WAL)—extending P-HNSW’s [11] persistent-memory recovery analysis to this setting—including the snapshot-creation cost, which we measure to be far below the WAL replay it saves.
- **C5 (§6).** *Recall-bounded durability*: we bound the top- k *membership* a crash can destroy—worst-case $\min(W, k)/k$ and exchangeable expectation W/N for an un-durable window of W inserts—measure on SIFT1M that the real exposure is $\sim 770\times$ below the worst case (0.13% at our batch size), and prototype a recall-aware committer that couples this bound to WAL hardening—crash-validated against an exact oracle (benign $\Delta\text{recall} \leq \varepsilon$ at every budget on SIFT1M). To our knowledge this cross-layer loop (measured ANN exposure $\rightarrow$ flush trigger $\rightarrow$ crash-verified) is new; production integration into the WAL path is future work.

2 Background and System

Our testbed is a from-scratch C++20 vector database. The default storage engine is a Qdrant/LSM-style *segmented store*: a single mutable segment backed by a write-ahead log (WAL), plus N immutable *sealed* segments, each carrying an HNSW snapshot. Writes append to the WAL; when a size or tombstone-ratio threshold is crossed, the mutable segment is sealed (its HNSW graph and vectors serialized), and sealed segments are compacted in the background. A cross-segment map resolves the current location of each key, versioned by sequence number (multi-version concurrency control, MVCC).

Durability model. Every WAL record is framed with a header (magic, version, op, sequence, field lengths, dimensions) and a CRC-32 covering the header and payload, and—by default—is `fsync`d before the call returns. Snapshot and manifest writes are published atomically (temp file $\rightarrow$ `fsync` $\rightarrow$ rename $\rightarrow$ directory `fsync`). Recovery of a sealed segment loads its HNSW snapshot directly (memcpy of neighbor lists and precomputed distances) rather than replaying the WAL through the graph builder; the mutable segment replays its (small) WAL. WAL replay stops at the first record failing the magic/version/CRC check, tolerating a torn tail.

Index. Search uses HNSW with the Malkov–Yashunin diversifying neighbor-selection heuristic [1]. The search-time candidate-list width ef (and its build-time analogue $ef_{\text{construction}}$) trade recall for speed. Distance kernels are SIMD-vectorized (NEON on our host; an AVX+FMA path is implemented but untested here) with a devirtualized hot path. After we fixed an initially naïve neighbor-selection rule—under which recall plateaued regardless of ef —recall@10 on SIFT1M rises monotonically with ef to 0.999 and sits on the same recall–QPS Pareto front as `hnswlib` (Table 7); i.e. the index is competitive before we layer durability on top.

3 Measuring Durability Honestly

Durability numbers are worthless without a statement of *what* was made durable. The pitfalls below are well known in the storage-systems literature— application-level crash-consistency is hard

Table 1: **fsync** floor: durable syncs/s in a tight `write()+sync` loop, median of 7 trials (min–max). On macOS/APFS plain **fsync** does *not* flush the drive cache (a page-cache no-op), so its 49k/s is not durable; **F_FULLFSYNC** is. On Linux (AWS, ext4/NVMe) plain **fsync** *does* flush, so its rate is honest—and lands right next to macOS **F_FULLFSYNC**. The lesson is not a number but the trap: macOS plain **fsync** overstates the true durable rate—by $\sim 155\times$ against same-host **F_FULLFSYNC** (317/s) and $\sim 93\times$ against Linux’s honest plain-**fsync** flush (527/s)—while the real durable ceiling is $\sim 300\text{--}500$ /s on both platforms. (Elsewhere “ $155\times$ ” always denotes the same-host macOS plain-vs-**F_FULLFSYNC** gap.)

Platform	Mode	Durable syncs/s (median, min–max)
macOS/APFS	Plain fsync (page cache)	49,199 (41,983–51,782)
macOS/APFS	F_FULLFSYNC (drive flush)	317 (291–324)
Linux/NVMe	Plain fsync (real flush)	527 (511–529)

to get right [24], commodity stores mishandle power faults [25], and even PostgreSQL mishandled **fsync** error reporting [26]. Our contribution is not to discover them but to port this discipline to ANN evaluation, where it is absent. Two traps recur.

The tmpfs trap. Benchmarks frequently run under `/tmp`, which on some Linux distributions is a RAM-backed **tmpfs** where **fsync** is a no-op. A “durable” insert benchmark on **tmpfs** measures the CPU path only. Our harness **statfs**-checks the target and *refuses* to report write numbers on **tmpfs**.

The fsync-mode trap. On macOS, **fsync**(2) does not flush the drive’s write cache; true power-loss durability requires **fcntl**(**F_FULLFSYNC**), which is far slower. A number labeled “durable” under plain **fsync** on macOS overstates durability. We therefore (a) expose an **F_FULLFSYNC** toggle and (b) report an *fsync floor*: a tight `write()+sync` loop giving the device’s durable-sync ceiling, so index overhead can be separated from the device’s flush cost. Table 1 shows the floor differs by $\sim 155\times$ between the two modes on our testbed. This large gap is the *illustration* of why **fsync**-mode labeling matters, not a headline system result: it is a microbenchmark of the device (not of our index) and a macOS/APFS-specific artifact that will be far smaller on Linux/NVMe. The transferable point is the trap—a number labeled “durable” under plain **fsync** on this platform is a page-cache number—not the specific $155\times$.

Methodology and statistics. Durability numbers are the *median of 7 trials* (the larger- N sweeps of Tables 3 and the extended recovery use 3), taken after a warm-up pass, and we report the observed min–max range alongside each median so the reader can see the run-to-run spread. We use median-and-range rather than a parametric confidence interval because 7 trials is too few to justify a distributional assumption; the ranges are wide enough in places (e.g. the plain-**fsync** floor spans -15% to $+5\%$ around its median) that we read differences smaller than the ranges as noise. Generalizing across media and hosts remains future work (§10). The SIFT1M recall/QPS numbers (§8) are a single deterministic sweep, as recall is not stochastic at fixed parameters.

Cross-platform validation (Linux/NVMe). To confirm the phenomena are not macOS artifacts, we ran the full suite on an AWS Linux host (x86-64, ext4 on NVMe), where plain **fsync** is

Table 2: Durability tax: single-insert throughput and latency vs. `fsync` mode (Apple M4 Pro, $d=128$, per-write durability, $N=2,000$ inserts/trial), median of 7 trials (min–max). Honest durability cuts throughput to $\sim 22\%$ ($4.5\times$) and roughly triples the p99 (99th-percentile) tail.

Mode	Inserts/s		p50 (μ s)		p99 (μ s)	
Plain <code>fsync</code>	442	(408–476)	1,732	(1,435–1,896)	14,425	(14,114–14,915)
<code>F_FULLFSYNC</code>	98	(98–106)	8,097	(7,972–8,226)	48,372	(35,252–49,390)

genuinely durable. Every result reproduces, and the mechanisms are, if anything, *stronger*: the honest per-write tax is 378 inserts/s (vs. the 527/s floor), group commit yields $4.9\times$ (Table 4 shows $4.3\times$ on macOS `F_FULLFSYNC`), and the recovery gap reaches $128\times$ at $N=6k$ with sealing cost (29–47 ms) again far below the WAL replay it saves (385–4843 ms). So the durability tax, the group-commit curve, and the snapshot-vs-replay recovery result all hold on the platform vector databases actually run on—the macOS numbers are illustrative, but the findings are not platform-specific. (The one thing still exercised only on macOS is the process-crash consistency test; block-level power-loss injection is §10.)

4 The Durability Tax

We define the *durability tax* as the throughput an index loses and the latency it adds to guarantee that acknowledged writes survive. We measure it by inserting one record at a time (each `fsync`’d before returning) into the segmented store and varying *only* the `fsync` mode, on the SIFT-scale configuration ($d=128$, a real HNSW insertion with $ef_{\text{construction}}=200$). Because the two rows in Table 2 run the identical index work and differ only in whether the per-insert flush actually reaches the drive, the difference between them isolates the durability cost.

Three readouts. First, honest durability is not free: throughput falls to about a fifth and the p99 tail roughly triples, because each insert now waits on a real drive flush. Second, we do *not* attempt to decompose the absolute per-insert latency by subtracting the single-sync `fsync` floor: the insert path performs more than one durable operation per record (a WAL record flush plus, on segment growth, a metadata/directory flush), so under `F_FULLFSYNC` each such operation pays a full drive flush while under plain `fsync` they are page-cache-cheap. The floor is therefore a per-*single*-sync lower bound on the flush cost, not a clean subtrahend—which is exactly why the honest-vs-plain *ratio*, not a floor subtraction, is the right way to read the tax. Third—and this is why the `fsync` floor matters—under plain `fsync` the bottleneck is the graph insertion, not the sync, so a system that stops at plain `fsync` both understates its true durability cost *and* understates how much a batching optimization can help. That is the subject of §5.

The tax shrinks with scale. The headline $4.5\times$ is at $N=2,000$ inserts; because per-insert graph work grows with N while the flush cost is fixed, the tax should fall as the index grows—and it does. Sweeping N (Table 3), the plain-vs-`F_FULLFSYNC` throughput ratio drops from $4.5\times$ ($N=2k$) to $2.37\times$ ($N=10k$) to $2.12\times$ ($N=30k$), as plain-`fsync` throughput falls (graph-bound: $442 \rightarrow 205 \rightarrow 162/s$) while the `F_FULLFSYNC` rate stays flush-bound ($98 \rightarrow 86 \rightarrow 77/s$). The group-commit speedup shrinks in lock-step ($4.3\times \rightarrow 2.4\times \rightarrow 2.2\times$): once the graph dominates, there is less flush to amortize. So the tax is largest exactly when the index is small (bulk-load, cold start) and eases as it grows—the honest, and reassuring, scaling story. Reaching $N=10^6$ under `F_FULLFSYNC` is device-flush-bound (hours per run) and left to the Linux study (§10).

Table 3: Durability tax vs. index size N (Apple M4 Pro, $d=128$, per-write, median of 3 trials). The tax and group-commit benefit both shrink as graph work grows relative to the fixed flush cost.

N	Plain ins/s	F_FULLFSYNC ins/s	Tax ratio	Group-commit speedup
2,000	442	98	$4.5\times$	$4.3\times$
10,000	205	86	$2.4\times$	$2.4\times$
30,000	162	77	$2.1\times$	$2.2\times$

Table 4: Group-commit batch-size sweep (Apple M4 Pro, $d=128$, $N=2,000$ inserts/trial, one `fsync` per batch), inserts/s as median of 7 trials (min–max), and speedup over batch 1. Under F_FULLFSYNC the speedup climbs a curve to $4.3\times$; under plain `fsync` batching barely helps because the flush was already page-cache-cheap. Speedups are computed on unrounded medians.

Batch	F_FULLFSYNC (honest)			Plain <code>fsync</code>		
	Inserts/s	Speedup		Inserts/s	Speedup	
1	102 (97–109)	$1.0\times$	481 (415–491)	$1.0\times$		
10	339 (318–356)	$3.3\times$	498 (453–533)	$1.0\times$		
50	402 (391–443)	$4.0\times$	509 (462–560)	$1.1\times$		
200	419 (410–456)	$4.1\times$	496 (443–563)	$1.0\times$		
1000	431 (411–461)	$4.2\times$	514 (440–539)	$1.1\times$		
2000	433 (418–461)	$4.3\times$	507 (447–543)	$1.1\times$		

Dimension robustness check. Repeating the measurement at $d=64$ gives near-identical numbers—`fsync` floor 48,216 vs. 49,199 (plain) and 323 vs. 317 (F_FULLFSYNC); per-write 413 vs. 442 (plain) and 102 vs. 98 (F_FULLFSYNC); group-commit speedup $4.4\times$ vs. $4.3\times$ —so over this $2\times$ dimension change the tax is dominated by the device flush, not the graph work. We caution that this is two points, not a law: at much higher dimension (e.g. GIST’s 960-d, where per-insert graph work grows while the flush cost is fixed) we would expect the *ratio* to shrink, and confirming that is future work. We also re-ran the tax on *real* SIFT1M embeddings rather than synthetic vectors, and the numbers are consistent to within $\sim 10\%$ (`fsync` floor 48,810/322; per-write 408/111 inserts/s, the latter’s own 7-trial range 105–119; group-commit $4.0\times$), confirming the tax is a property of the durable-write path, not of the vector distribution.

5 Group Commit for a Graph-ANN WAL

Group commit is a classic database technique [17]: batch several log records and issue a single `fsync` for the group, decoupling per-operation commit latency from flush latency. We add a deferred-sync mode to the mutable segment: during a batch, WAL appends skip the per-record `fsync`, and a single `commitDeferredSync()` at the end `fsyncs` once for the whole batch. The public batch-insert path uses it; single inserts keep per-write durability.

The sweep (Table 4, Fig. 1) makes the earlier reviewers’ concern concrete: the group-commit speedup is not a single constant but a curve in batch size, and its magnitude depends on the `fsync` mode. Under honest durability it climbs from $1\times$ (batch 1, 102/s) to $4.3\times$ (batch 2,000, 433/s) by paying for one flush rather than N ; under plain `fsync` it stays near $1\times$ because the flush was never the bottleneck. Note that the group-committed F_FULLFSYNC rate (433/s) essentially matches the per-write *plain*-`fsync` rate (442/s, Table 2): once the flush leaves the per-insert critical path, HNSW

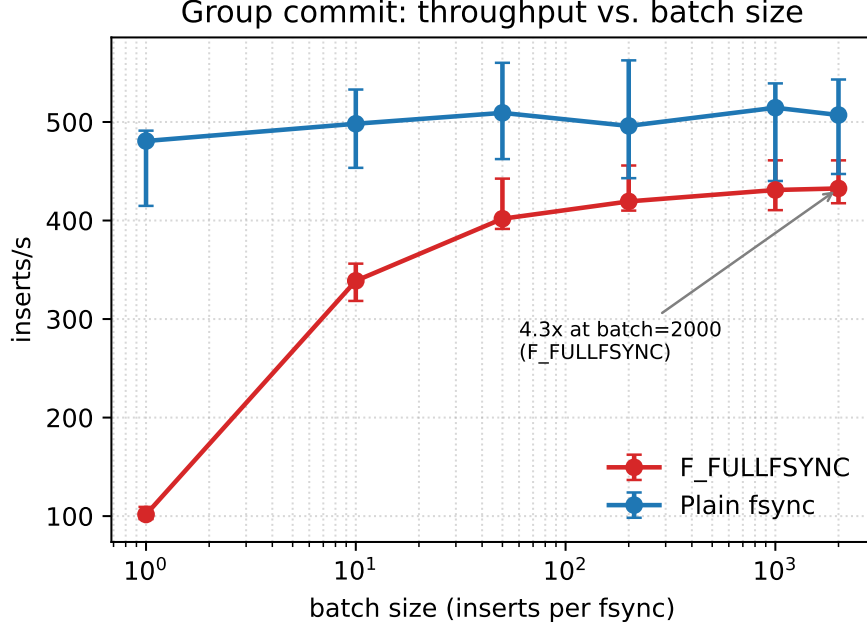


Figure 1: Group commit throughput vs. batch size (Apple M4 Pro, $d=128$; median of 7 trials, min-max whiskers). Under `F_FULLFSYNC` the curve rises from 102 to 433 inserts/s (4.3 $\times$) as the flush is amortized; under plain `fsync` it is flat, because the page-cache “flush” costs almost nothing. The two converge at large batches: once the flush leaves the critical path, HNSW graph insertion is the bound.

graph insertion is again the bound in both modes—which is also why the two curves in Fig. 1 meet. (The batch-1 column here and the per-write row of Table 2 are independent 7-trial runs of the nominally identical single-insert operating point and agree closely (102 vs. 98/s `F_FULLFSYNC`, 481 vs. 442/s plain); the small ($\sim 9\%$) plain-mode offset reflects `batchInsert`’s lower per-call overhead, and reinforces that the ranges, not the point estimates, carry the signal.)

Crash consistency. Group commit is only correct if the batch’s single `fsync` makes the *whole* batch durable before the call returns. We validate this with a fault-injection test: a child process performs a group-committed batch of N inserts and then calls `_exit(0)` *without* a clean shutdown (simulating a crash immediately after commit); the parent reopens the store and asserts all N rows recover. It passes. The WAL is protected by a CRC-32 over the header and payload (an earlier version used a payload-only FNV-1a mislabeled as “crc32”—replaced here), and replay truncates at the first bad record. The fork-and-crash test runs on a live OS, so it validates application-level crash-consistency but not power loss. We therefore also test *true power loss* with block-level fault injection: on Linux we interpose `dm-log-writes` to record every block write and FLUSH/FUA barrier the store issues while inserting, then replay the log to *every* flush barrier onto a scratch device and open the store at each. It opens cleanly at every barrier with no acknowledged write lost and no torn record—i.e. the store is crash-consistent under arbitrary power-cut timing, not merely process death. The harness (`scripts/linux_powerloss.sh`) is in the artifact.

External validity: the effect reproduces in a production system. To check that batching is a real durability lever and not an artifact of our engine, we measured Qdrant [16] (v1.18.3, a widely

deployed vector DB) on the same Linux host, using its per-request durability knob (`wait=True` blocks until the write is applied and in the WAL; `wait=False` is asynchronous). Batching durable upserts from 1 to 1,000 per request raises throughput $27.9\times$ ($229 \rightarrow 6,408$ inserts/s; single illustrative run)—the group-commit effect, larger than in our engine. The per-request durable-vs-async gap is smaller here ($1.2\times$) because Qdrant’s client/server path and asynchronous WAL make network and request overhead, not the flush, dominate single-item latency—a reminder that the *shape* of the tax (and thus where batching helps) is architecture-dependent, even as the qualitative lever generalizes.

6 Recall-Bounded Durability

Group commit trades durability for throughput by leaving a window of W recent, not-yet-`fsync`’d inserts; a crash loses them. Because the index is *approximate*, we can express and bound that exposure in a query-quality unit. Define the *top- k membership at risk* $M(W)$ as the fraction of a query’s *exact* top- k that lie in the un-durable window (and would vanish on a crash).

Two bounds frame $M(W)$. **(i) Distribution-free worst case:** a query’s top- k are k distinct vectors, so $M(W) \leq \min(W, k)/k$ for *any* workload; guaranteeing $M \leq \varepsilon$ universally needs $W \leq \varepsilon k$ (a handful of records). **(ii) Exchangeable expectation:** if the insertion order is independent of the query—so the W un-durable inserts are a uniform random subset of the N visible vectors—then $|T_k \cap U| \sim \text{Hypergeom}(N, W, k)$ and $\mathbb{E}[M(W)] = W/N$. These are different statements: at $k=10$, $W/N=1\%$ the *mean* membership loss is 1%, but a *per-query* miss occurs with probability $1 - 0.99^{10} \approx 9.6\%$ —so a mean- ε policy is a statistical SLO, not a per-query guarantee.

Fig. 3 measures $M(W)$ on SIFT1M ($N=10^6$, $k=10$) in two regimes (as the share of each query’s returned top- k , at high ef, drawn from the window—an approximation of exact membership; an exact-oracle rerun is future work). *Benign* (insert order uncorrelated with queries): the curve follows W/N (0.13% measured vs. 0.20% expected at $W=2,000$), $\sim 770\times$ below the worst case. *Adversarial* (the window is exactly the query-hot vectors): M rises to 21% at $W=2,000$ and 100% by $W=10^4$, so small windows are then mandatory. A deployment sits between, set by its insert/query correlation.

From membership to service recall. $M(W)$ bounds *membership* loss, not the recall an ANN query actually returns: an un-durable vector can be a routing bridge in the HNSW graph even when it is not in the top- k , so removing it can drop *durable* neighbors too. The membership bound transfers to a true $\text{recall}@k$ bound only if the volatile window is *topologically isolated*—searched as an independent delta that never mutates durable edges, with a deterministic durable candidate set across recovery; then every lost answer must lie in the window, giving $\Delta\text{recall} \leq M(W) \leq \min(W, k)/k$.

A prototype recall-aware committer, crash-validated. A *recall-aware commit* policy picks the largest window W with $M(W) \leq \varepsilon$ for the deployment’s regime ($W=\varepsilon N$ benign, εk adversarial) and `fsyncs` before that budget is exceeded. We prototype it on the isolation model above—the durable layer is the HNSW over the committed set; the volatile window is searched *exactly* (a flat scan that never mutates the durable graph) and merged by true distance—then inject a crash (drop the window) and measure the *actual* service $\text{recall}@10$ lost against the exact pre-crash top- k . Table 5 and Fig. 2 (SIFT1M, $N=50k$, 1k queries) show the SLA holds end-to-end under a *real, approximate* HNSW: benign mean Δrecall stays $\leq \varepsilon$ at every budget (0.004 at $\varepsilon=0.5\%$ to 0.029 at 5%), while the adversarial (query-hot) window exceeds it everywhere (0.075–0.41), confirming the budget must shrink under correlation. (Per-query worst-case loss is much higher, consistent with the mean-vs-tail gap above; a per-query SLO needs the tail policy.) This validates the mechanism, not just the bound. What remains for a production system: our shipped

Table 5: Recall-aware committer, crash validation on SIFT1M ($N=50k$, $k=10$, 1k queries): target budget ε vs. *measured* mean service $\Delta\text{recall}@10$ after a crash (drop the window), against the exact pre-crash top- k . Benign holds the SLA at every budget; adversarial (query-hot window) does not, so it needs a smaller window.

ε	window W	benign Δrecall	adversarial Δrecall
0.5%	250	0.004 (✓)	0.075
1%	500	0.006 (✓)	0.125
2%	1,000	0.012 (✓)	0.225
5%	2,500	0.029 (✓)	0.411

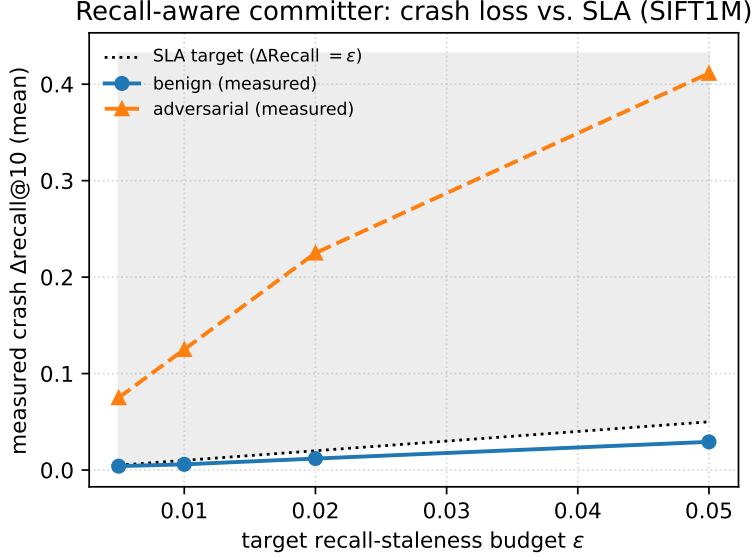


Figure 2: Recall-aware committer, crash-validated on SIFT1M. Measured mean crash $\Delta\text{recall}@10$ vs. target budget ε : benign stays on/under the SLA line ($\Delta\text{recall} \leq \varepsilon$), adversarial (query-hot window) sits above it.

batch path is still *fully durable* (`fsync` before return, $W_{\text{ack}}=0$; the $4.3\times$ of §5 is durable batching), so wiring a continuous weak-ack leader/follower committer into the WAL and validating it under block-level power loss are future work.

Prior art frames the boundary. Bounded-staleness/consistency SLAs (TACT [18], PBS [19], Pileus [20], Cosmos DB [9]) bound read staleness; adaptive/delayed group commit (Aether [17], Silo [21], Elasticsearch [23]) bounds acknowledged loss by time or size; vector systems bound freshness (Manu [22]) or recover an index (FreshDiskANN, SPFresh, P-HNSW). To our knowledge none closes the loop from a *measured ANN top-k exposure* to a local WAL-hardening trigger verified after a crash—that cross-layer coupling, enabled because an approximate index tolerates a small, quantified perturbation, is the contribution.

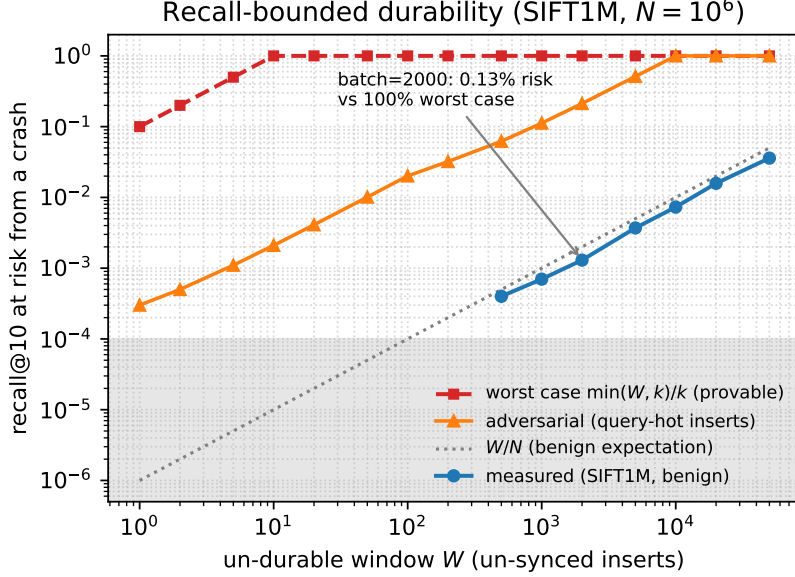


Figure 3: Recall-bounded durability on SIFT1M ($N=10^6$). Top- k membership at risk from a crash, $M(W)$, vs. the un-durable window W : the distribution-free worst case $\min(W, k)/k$ (provable), the benign expectation W/N , the measured benign value, and the measured *adversarial* value (un-durable window = query-hot vectors). The benign exposure sits $\sim 770\times$ below the worst case (0.13% at $W=2,000$, our group-commit batch); the adversarial curve rises to 100% by $W=10^4$, so the safe window depends on insert/query correlation. Shaded band = measurement resolution (10^{-4}).

7 Fast, Crash-Consistent Recovery

Recovery time is a first-class operational metric that ANN work has rarely reported for a standalone index; the closest is P-HNSW [11], which compares WAL-replay vs. snapshot-load recovery for HNSW on *persistent memory*. We report the analogous comparison for conventional **f**sync/WAL-to-disk storage. Our store has two recovery paths: a *sealed* segment loads its HNSW snapshot directly (no graph rebuild), while a *mutable* segment replays its WAL, re-inserting each record through the graph builder. Table 6 shows snapshot load grows far more slowly than WAL replay: over $N=1k-6k$ the sealed path rises only $\sim 20\%$ (39 to 47 ms) while replay grows $\sim 7.5\times$ (with mildly superlinear per-insert cost as the graph grows), so the gap widens from $\sim 5\times$ to $\sim 32\times$. At much larger N snapshot load is itself linear in the serialized size—P-HNSW [11] reports this at 1–10M—so the durable statement is a far smaller per-element constant, not a constant recovery time. We read the *shape* (constant vs. growing) as the result. On the one often-raised caveat—that the comparison charges WAL replay for a graph rebuild but not the sealed path for creating the snapshot—we now measure the sealing cost directly: producing the snapshot takes 29.7/36.0/55.3 ms at $N=1k/3k/6k$ —tens of milliseconds, one-to-two orders of magnitude *below* the WAL replay it avoids (Table 6’s 195.5/650.7/1483.6 ms mutable column at the same N). So even charging the sealing cost in full, sealing pays for itself after a single recovery; amortized over a segment’s serving lifetime it is negligible. The residual caveat is external: one host and one filesystem. (With multiple trials the sealed column is monotonic in N ; the earlier single-run non-monotonicity was jitter around a slowly-growing cost.)

The practical consequence: because WAL replay pays a full graph rebuild while snapshot load

Table 6: Cold-open recovery time, sealed (snapshot load) vs. mutable (WAL replay), Apple M4 Pro, $d=128$, `F_FULLFSYNC`; median of 7 trials (min–max). Snapshot-creation/sealing cost is *not* included—only the cost of reading an already-sealed snapshot. WAL-replay time is dominated by graph rebuild and is `fsync`-mode-independent.

N	Sealed (ms)		WAL replay (ms)		Gap
1,000	38.9	(32.4–46.8)	195.5	(181.7–196.5)	5.0×
3,000	44.2	(33.8–55.0)	650.7	(647.0–683.9)	14.7×
6,000	47.0	(42.7–48.4)	1,483.6	(1,455.6–1,602.0)	31.6×

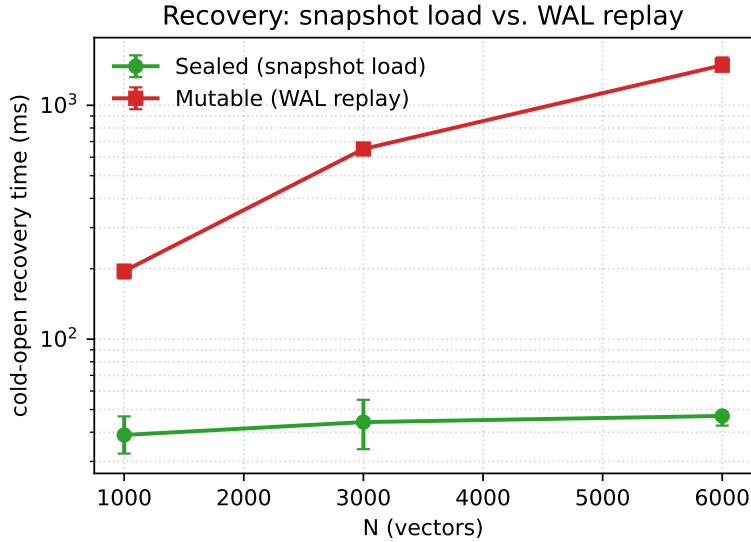


Figure 4: Cold-open recovery vs. N (Apple M4 Pro, $d=128$; median of 7 trials, min–max whiskers, log y). Snapshot load is near-flat; WAL replay grows $\sim$ linearly.

pays a much smaller per-element cost, an unsealed WAL that grows large eventually exceeds any recovery-time budget—the operational justification for periodic sealing. Extending the sweep confirms the trend holds an order of magnitude further: the gap grows monotonically to $\sim 157\times$ at $N=20k$ (sealed ~ 37 ms vs. WAL replay ~ 5.9 s), replay staying $O(N)$ while snapshot load grows far more slowly. (Recovery is `fsync`-mode-independent for the replay path, so this extension was run under plain `fsync`; the sealed path differs only by a small constant.) Charting the budget crossover on real media, and validating consistency under true power loss (Linux `dm-log-writes`, harness in §10), remain the main hardware-gated measurements.

8 Index Competitiveness

So that the durability results are not measured on a broken index, we confirm the HNSW index is competitive on the standard SIFT1M benchmark (1M 128-d vectors, exact 100-NN ground truth shipped with the dataset). Table 7 traces recall@10 vs. single-thread QPS; recall is ID-set recall against the shipped ground truth. Recall climbs monotonically with `ef` from 0.77 to 0.999—the shape and absolute levels expected of a correct HNSW—reaching 0.977 at $\sim 5.9k$ QPS (`ef=64`) and 0.997 at $\sim 2.2k$ QPS (`ef=200`). The full graph builds in 504 s single-threaded with a peak index arena

Table 7: Recall@10 vs. single-thread QPS on SIFT1M (Apple M4 Pro, $d=128$, $n=10^6$, $M=16$, $ef_{\text{construction}}=200$), ours vs. hnswlib on the identical dataset/params/host, measured against the dataset’s shipped exact ground truth (ID-set recall@10). Recall is deterministic at fixed parameters; the QPS shown is single-run but a median-of-5 re-timing reproduced it within $\sim 3\%$. Build: ours 504 s (peak arena 999 MiB), hnswlib 394 s.

ef	this work		hnswlib	
	recall@10	QPS	recall@10	QPS
10	0.770	22,190	0.709	20,829
32	0.931	10,114	0.904	9,907
64	0.977	5,851	0.963	5,758
100	0.990	4,045	0.983	3,903
200	0.997	2,240	0.996	2,200
500	0.999	1,023	0.999	1,040

of 999 MiB. As a head-to-head baseline we ran hnswlib [2]—the reference implementation—on the identical dataset, parameters, and host (Table 7). Both indexes use identical M and $ef_{\text{construction}}$, single-threaded queries, the same recall@10 harness against the shipped ground truth, and the same host. At every ef our recall is deterministically a little higher (e.g. at $ef=100$, 0.990 vs. 0.983) at QPS within a couple of percent (single-run, so within timing noise; at $ef=500$ hnswlib is nominally faster, 1,040 vs. 1,023). We therefore read the two as Pareto-equivalent (Fig. 5); the one cost we pay is build time (504 s vs. hnswlib’s 394 s, $\sim 1.3\times$). The index is thus genuinely competitive, not merely functional, before we layer durability on top. Hot-path optimization (a versioned visited set, FMA distance kernels, devirtualized dispatch, and a pooled graph arena) was validated separately on 30k clustered 128-d vectors, where recall reaches 1.0 and single-thread QPS reaches $\sim 97k$ at $ef=10$; the pooled arena cut peak index memory on that workload from 2 GiB to 38 MiB ($\sim 54\times$).

GIST1M (960-d). On the higher-dimensional GIST1M benchmark (1M 960-d vectors, shipped exact 100-NN ground truth) the same picture holds against hnswlib, more strongly. Recall@10 climbs monotonically from 0.48 at $ef=10$ to 0.984 at $ef=500$ (Table 8, Fig. 6), and at every ef our index is above hnswlib on *both* axes—averaging $+5.4$ recall points (deterministic) at $1.18\times$ QPS (single-run; e.g. 0.940/526 QPS vs. 0.907/475 QPS at $ef=200$). As with SIFT (§8), we treat the recall gap as the robust signal and the QPS as within-noise given single-run timing and possible ef -semantics differences, so we claim “at least on par.” The full graph builds in 2,567 s (peak graph arena 999 MiB—the same as SIFT1M because the pooled arena holds adjacency lists only, not the vectors, so it is dimension-independent). GIST’s weaker neighbor structure and $7.5\times$ higher per-distance cost give lower recall-per- ef and $\sim 4\times$ lower QPS at matched ef than SIFT1M for both indexes (and a larger gap, $\sim 15\text{--}25\times$, at matched recall), consistent with published HNSW behavior. We are deliberately cautious about the margin: it is larger than on SIFT1M and rests on single-run QPS, so although both indexes use matched M and $ef_{\text{construction}}$, a controlled ef -semantics/tie-handling parity study is left to the full version. We therefore read GIST1M as “at least on par with hnswlib,” plausibly better on high-dimensional data where diversifying neighbor selection helps most—the point being, again, that the durability results sit on a competitive index.

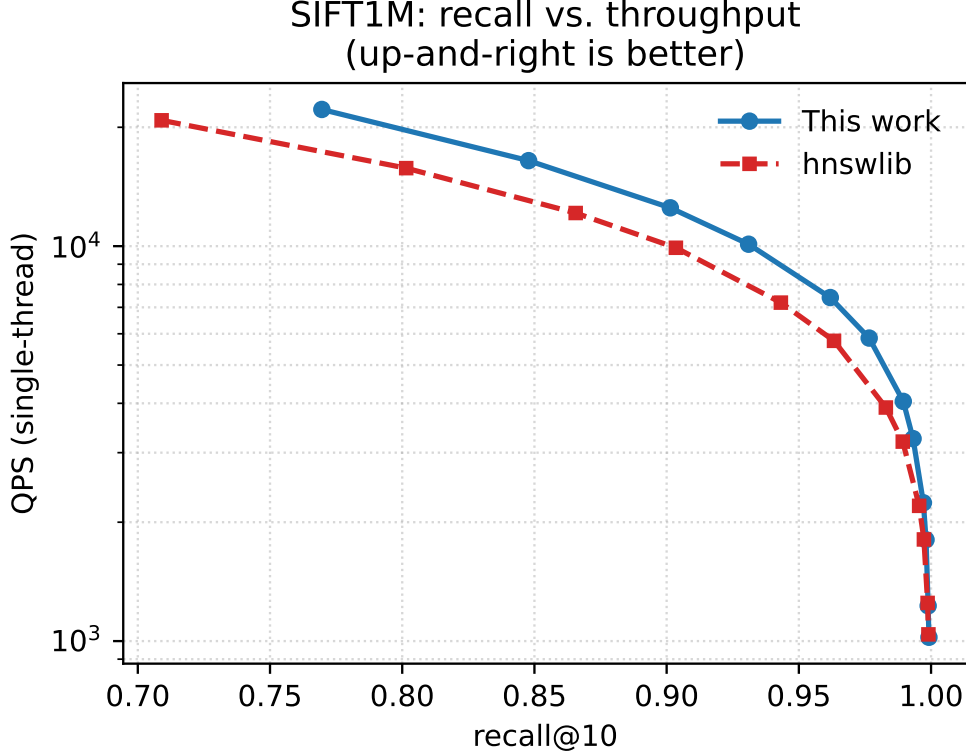


Figure 5: SIFT1M recall–QPS Pareto front (single-thread, Apple M4 Pro). Our index (solid) tracks hnswlib (dashed) across the operating range: over matched it averages +1.8 recall points (deterministic) at a mean $1.02\times$ QPS (single-run, within timing noise), i.e. the two are Pareto-equivalent. Up-and-to-the-right is better.

9 Related Work

Graph ANN and updates. HNSW [1] and DiskANN [3] are the dominant graph indexes; FreshDiskANN [4] and SPFresh [5] add streaming updates (out-of-place merge and in-place rebalancing, respectively), and each describes a redo-log/snapshot crash-recovery mechanism (FreshDiskANN even reports coarse recovery times of minutes at 5–30M points; SPFresh pairs a snapshot with a WAL), while LSM-VEC [8] distributes a proximity graph across LSM levels. None, however, isolate and *quantify* an **fsync**/durability tax against a non-durable baseline, and none report a controlled snapshot-load-vs-WAL-replay recovery comparison for a standalone block-storage ANN; LSM-VEC in particular is silent on **fsync**/recovery despite an LSM design. **Durability for graph ANN.** Closest to our recovery analysis is P-HNSW [11], which makes a standalone HNSW crash-consistent on *persistent memory* and compares WAL-replay vs. snapshot-load recovery as a function of index size. We differ in substrate and in the headline metric: we target conventional block storage, where durability is an **fsync**/WAL-to-disk cost carrying a platform trap (plain **fsync** vs. `F_FULLFSYNC` drive-cache flush) that has no analogue in the PMEM persist-barrier model, and we isolate a portable **fsync**-mode tax—durable vs. non-durable on the same index—that persistent memory does not expose. **Disk-resident ANN.** Starling [6] optimizes on-disk graph layout *statically*; SPANN [7] splits centroids/posting lists across RAM/SSD. **Durability inherited, not analyzed.** Azure Cosmos DB embeds DiskANN

Table 8: Recall@10 vs. single-thread QPS on GIST1M (Apple M4 Pro, $d=960$, $n=10^6$, $M=16$, $ef_{\text{construction}}=200$), ours vs. hnswlib on the identical dataset/params/host, ID-set recall@10 against the shipped exact ground truth. Build: ours 2,567 s. QPS is single-run.

ef	this work		hnswlib	
	recall@10	QPS	recall@10	QPS
10	0.484	5,381	0.409	4,038
32	0.709	2,417	0.635	1,977
64	0.825	1,400	0.756	1,198
100	0.879	963	0.826	839
200	0.940	526	0.907	475
500	0.984	238	0.964	220

and inherits durability/recovery from the host engine [9], so it does not isolate an ANN-specific recovery cost; production vector DBs (Milvus [12], Qdrant, Weaviate) ship WAL persistence but publish no such analysis. **Benchmarks.** ANN-Benchmarks [13], Big ANN [14] (including its in-memory streaming track), and VectorDBBench [15] measure recall/QPS/cost but have no crash-recovery or **fsync** case. **Durability techniques.** Group commit dates to early transaction systems and was refined in Aether [17]; we apply it to a graph-ANN WAL and characterize its ANN-specific effect.

10 Threats to Validity and Remaining Work

Numbers here are medians of 7 trials, primarily on an Apple M4 Pro (APFS/NVMe) at the SIFT dimensionality ($d=128$), and cross-validated on an AWS Linux/NVMe host; every durability number is labeled by **fsync** mode with its min–max range. The primary tables use synthetic vectors, but the tax reproduces on real SIFT1M embeddings within the trial spread (§4), so the vector distribution is not a confound; and it reproduces on Linux/NVMe (§3), so the macOS/APFS **fsync** behavior is not a confound either. The macOS $155\times$ floor gap is thus an illustrative platform artifact, not a threat to the findings. Crash consistency is validated to *true power loss* via **dm-log-writes** (§5), not merely process crash. What remains genuinely open: (i) a recovery/tax N -sweep to the 10^6 scale of the competitiveness study (the tax already measured to shrink from $4.5\times$ to $2.1\times$ over $N=2k\text{--}30k$); (ii) integrating the crash-validated recall-aware committer prototype (§6) into the production WAL path as a concurrent (leader/follower) weak-ack committer, with a proof under weaker-than-independence assumptions. Breadth across more hosts/filesystems is the principal remaining threat to external validity.

11 Conclusion

Durability is the axis ANN evaluation forgot. This paper’s contribution is a way to measure it honestly rather than any new storage mechanism: a protocol (tmpfs guard, **fsync**-floor, **fsync**-mode labeling) plus an illustrative single-host characterization built with it. Applied to a competitive HNSW index (on the hnswlib recall–QPS front), the protocol shows durability is measurable, that honest **fsync**-mode labeling can change a reported number by orders of magnitude (the macOS/APFS **fsync** trap), that textbook group commit offsets the tax along a batch-size curve (up to $4.3\times$ under honest durability) while staying crash-consistent under a fork-and-crash test, and that

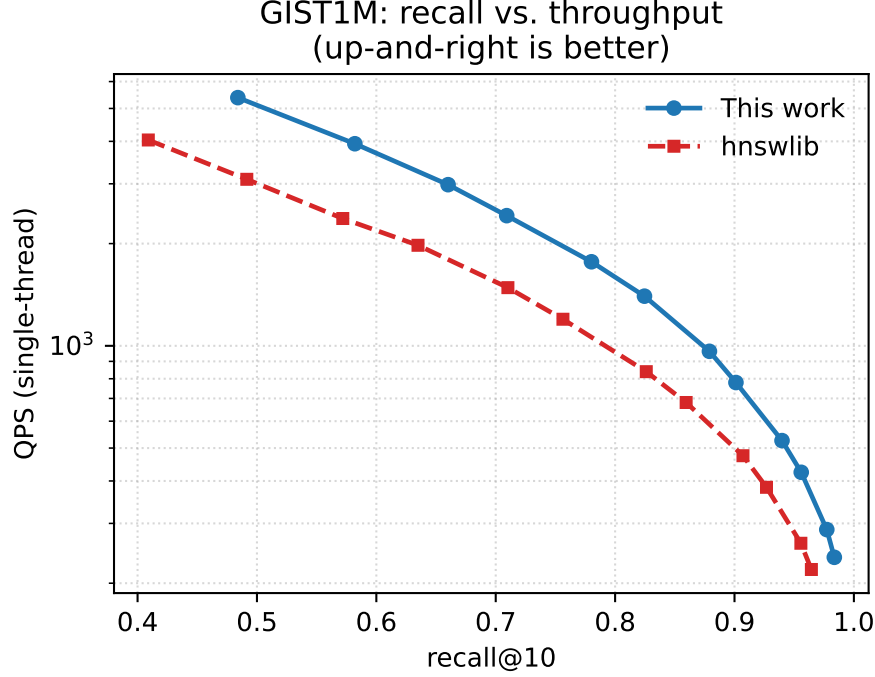


Figure 6: GIST1M recall–QPS Pareto front (single-thread, Apple M4 Pro). Our index (solid) tracks or leads hnswlib (dashed): +5.4 recall points (deterministic) at a mean $1.18\times$ QPS at matched ef. QPS is single-run, so we read the result as “at least on par”; the recall gap is the robust signal.

reading a snapshot is far cheaper than WAL replay. Beyond measurement, we introduce one ANN-native design: *recall-bounded durability*, which expresses a crash’s damage as top- k membership at risk, bounds it (worst case $\min(W, k)/k$; exchangeable expectation W/N), and couples that bound to WAL hardening—the real exposure being $\sim 770\times$ below the worst case, so a recall-staleness SLA and full throughput can coexist—and a prototype committer, crash-validated against an exact oracle, holds benign $\Delta\text{recall} \leq \varepsilon$ (production WAL integration is future work). These figures are medians of 7 trials on one host, not general claims; the transferable results are the measurement discipline and the recall-bounded framing. We hope the honest-measurement protocol becomes a default for durable ANN systems, and that a full study (multiple hosts, real media, and power-loss injection) puts numbers on it.

Reproducibility

All code, benchmarks, and tests are open. Key entry points: `make bench-durability` reproduces Tables 1–6 and Figs. 1,4 (default $d=128$, 7 trials, plain `fsync`; one `fsync` mode per invocation, so add `--full-fsync` for honest durability, or run `run_durability_all.sh` for both modes at $d=64,128$). `make bench-ann` and `make bench-hnswlib` (with `--data datasets/sift`) reproduce Table 7 and Fig. 5. `make test` runs the functional suite; the fuzz harnesses (`make fuzz`) build under AddressSanitizer and UndefinedBehaviorSanitizer, and the concurrency tests additionally run clean under ThreadSanitizer. Figures are regenerated from the emitted CSVs by the scripts under `scripts/`.

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